Incident handler's journal

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this journal as a way to log the key takeaways about the different cybersecurity tools or concepts you encounter in this course.

Date: 19/09/2026	**Entry:** #1
Description	Documenting a cybersecurity incident This incident occurred in the two phases: 1. **Detection and Analysis**: The scenario outlines how the organization first detected the ransomware incident. For the analysis step, the organization contacted several organizations for technical assistance. 2. **Containment, Eradication, and Recovery**: The scenario details some steps that the organization took to contain the incident. For example, the company shut down their computer systems. However, since they could not work to eradicate and recover from the incident alone, they contacted several other organizations for assistance.
Tool(s) used	None
The 5 W's	• **Who**: Unethical hackers that got access to the system. • **What**: A ransomware security incident • **Where**: At the healthcare clinic • **When**: Tuesday 9:00 a.m. • **Why**: The incident happened because unethical hackers were able to access the company's systems using a phishing attack. After gaining access, the attackers launched their ransomware on the company's systems, encrypting critical files. The attackers' motivation appears to be financial because the ransom note they left demanded a large sum of money in exchange for the decryption key.
Additional notes	1. Employees should be alerted to the dangers of phishing attacks. Additionally, verify suspicious communications before responding, report security incidents immediately to the response team, and conduct ongoing security awareness training.

Date: 21/09/2026	**Entry:** #2
Description	Analyzing a packet capture file
Tool(s) used	For this activity, I used Wireshark to analyze a packet capture file. Wireshark is a network protocol analyzer that uses a graphical user interface. The value of Wireshark in cybersecurity is that it allows security analysts to capture and analyze network traffic. This can help in detecting and investigating malicious activity.
The 5 W's	• **Who**: N/A • **What**: N/A • **Where**: N/A • **When**: N/A • **Why**: N/A
Additional notes	I've never used Wireshark before, so I was excited to begin this exercise and analyze a packet capture file. At first glance, the interface was very overwhelming. I can see why it's such a powerful tool for understanding network traffic.

Date: 27/09/2026	**Entry:** #3
Description	Capturing my first packet
Tool(s) used	For this activity, I used tcpdump to capture and analyze network traffic. Tcpdump is a network protocol analyzer that's accessed using the command-line interface. Similar to Wireshark, the value of tcpdump in cybersecurity is that it allows security analysts to capture, filter, and analyze network traffic.
The 5 W's	• **Who**: N/A • **What**: N/A

	• **Where**: N/A • **When**: N/A • **Why**: N/A
Additional notes	I'm still new to using the command-line interface, so using it to capture and filter network traffic was a challenge. I got stuck a couple of times because I used the wrong commands. But after carefully following the instructions and redoing some steps, I was able to get through this activity and capture network traffic.

Date: 01/10/2026	**Entry:** #4
Description	Investigate a suspicious file hash
Tool(s) used	For this activity, I used VirusTotal, which is an investigative tool that analyzes files and URLs for malicious content such as viruses, worms, trojans, and more. It's a very helpful tool to use if you want to quickly check if an indicator of compromise like a website or file has been reported as malicious by others in the cybersecurity community. For this activity, I used VirusTotal to analyze a file hash, which was reported as malicious. This incident occurred in the **Detection and Analysis** phase. The scenario put me in the place of a security analyst at a SOC investigating a suspicious file hash. After the suspicious file was detected by the security systems in place, I had to perform deeper analysis and investigation to determine if the alert signified a real threat.
The 5 W's	• **Who**: An unknown malicious actor • **What**: An email sent to an employee contained a malicious file attachment with the SHA-256 file hash of 54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b • **Where**: An employee's computer at a financial services company • **When**: At 1:20 p.m., an alert was sent to the organization's SOC after the intrusion detection system detected the file • **Why**: An employee was able to download and execute a malicious file attachment via e-mail.

Additional notes	We should consider improving security awareness training so that employees are careful with what they click on.

Date: 07/10/2026	**Entry:** #5
Description	Analyze logs with suricata
Tool(s) used	For this activity, I used Suricata, an open-source network analysis and threat detection engine that can function as an Intrusion Detection System (IDS) and Intrusion Prevention System (IPS). Suricata uses rules to examine network traffic, trigger alerts, and log events. It can be executed via the command-line interface to analyze packet capture (.pcap) files against custom rule files.
The 5 W's	• **Who**: N/A • **What**: N/A • **Where**: N/A • **When**: N/A • **Why**: N/A
Additional notes	During this activity, I explored and created custom rules in Suricata and tested them against network traffic data using a sample.pcap file. When running Suricata, it evaluates the rules against the traffic and generates alerts and log outputs in the /var/log/suricata directory: • **fast.log**: Contains quick alert entries generated whenever rule conditions are met. While considered a deprecated format for threat hunting, it is useful for quick checks and quality assurance during rule testing. **eve.json**: The primary and default event log format for Suricata. It records detailed JSON-formatted information on triggered alerts and network telemetry.

Reflections/Notes:

1. **Were there any specific activities that were challenging for you? Why or why not?**

Yes, working with command-line interface tools like tcpdump was quite a challenge at first due to syntax errors and unfamiliarity with the commands. Similarly, the Wireshark interface and rule analysis in Suricata were initially overwhelming because of the sheer amount of data and information displayed, though I managed to master them through guided practice.

2. **Has your understanding of incident detection and response changed after taking this course?**

Yes, my perspective has changed significantly. I now have a structured understanding of the various phases of the response cycle—such as Detection, Analysis, Containment, Eradication, and Recovery. Furthermore, I grasped the importance of complementing internal analysis with specialized tools like VirusTotal for investigating malicious hashes and Suricata for rule-based threat detection.

3. **Was there a specific tool or concept that you enjoyed the most? Why?**

The tool I enjoyed the most was VirusTotal, due to the speed and clarity with which it allows for verifying indicators of compromise and analyzing suspicious files within the community. I also found it fascinating to work with Wireshark and Suricata, as they enable detailed inspection of network traffic and an understanding of actual packet behavior.